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Seeing the Process Through the Worker's Eyes: Event Logs from Egocentric Factory Videos

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Abstract

Much of manual factory work remains invisible to information systems: moving materials, preparing tools, checking surfaces, and repairing defects can leave no event record. Egocentric video brings this work into view, but turning what a camera sees into a process log requires decisions about actions, names, and time. We present a construction method and linked data resource that make these decisions inspectable. A vision-language model produces timestamped descriptions of visible work, which we call generated transcripts. Pretrained models summarize these descriptions, derive shared factory labels, and extract labeled event intervals. The resulting logs support exploration of work composition, local sequences, and differences between recorded worker profiles. Worked examples show how the representation shapes the findings. Transport remains in the event log, but its duration is counted under other categories because the category rule uses broader process labels. A described episode can also acquire an additional event boundary during extraction. Structural checks and an author audit assess the records and their consistency with generated transcripts, while accuracy against the source videos remains an open question. Preserving the intermediate descriptions alongside the logs lets researchers trace measurements to their textual basis and investigate alternative representations of manual work.

Keywords: process mining; inferred event logs; egocentric video; measurement; data quality; manufacturing; foundation models

1. Introduction

Process mining uses timestamped event records to study operational processes [1]. Many manual factory tasks leave no such records. An information system may record an order or machine checkpoint while missing how a worker loads parts, moves material, checks a component, or switches tools. Video can make some of this work observable, but it does not directly provide the case identifiers, labels, and event boundaries that process mining needs [2].

Egocentric video records work from the camera wearer's perspective. The public Egocentric-10K resource provides factory recordings of this kind [3]. Figure 1 illustrates the viewpoint. Turning recordings into event logs requires decisions about what constitutes an event, which names to use, and how to interpret time. These are *measurement decisions* as well as extraction tasks. A well-formed log can still misrepresent the work if a behavior is omitted, a boundary is estimated poorly, or an analytical category hides a recorded action.

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Figure 1. Example frames from Build AI's Egocentric-10K resource [3]. They illustrate the camera viewpoint, rather than the accuracy of the derived event labels.

Video-to-log construction already includes language-model-based methods. Vid2log uses a pretrained vision-language model to interpret video and construct event data [4]. First-person workflow extraction also predates this study [5,6]. The question here is how to build a *shared event representation* from the available factory recordings and explain its consequences for process analysis. In particular, naming an activity, assigning an episode to that name, and mapping the label into a measurement category are separate decisions. A shared vocabulary makes within-factory comparison possible, but does not by itself make the resulting measurements accurate or comparable across tasks.

We study a *five-stage pipeline*: describe each recording, summarize each worker's recordings, derive shared factory labels, extract labeled event intervals, and combine the events into logs. The timestamped descriptions of visible actions are called *generated transcripts*. *Vocabulary induction* means deriving shared activity and process label lists from the recorded workers in each factory. These lists guide event extraction from the generated transcripts. The application covers six factories, 59 worker identifiers, and 322 recordings. The resulting 10,698 events are *inferred operational episodes*. They do not provide a complete census of production cycles or manufactured items.

Figure 2 outlines the path from recordings to analysis that guides the paper. After reviewing related work in Section 2, we define the events and explain their construction in Sections 3 and 4. Section 5 presents the released data. Sections 6 and 7 describe the assessment and report the analyses and factory cases. Sections 8 and 9 explain how the construction choices affect the findings and discuss the remaining limits.

The paper makes three contributions:

- **A construction design.** We specify how descriptive narration, aggregation across workers, factory vocabulary induction, and vocabulary-constrained extraction produce interval events. The foundation-model parameters remain fixed; the representation adapts to the available factory corpus. Formal definitions and implementation checks make the event and case choices explicit.

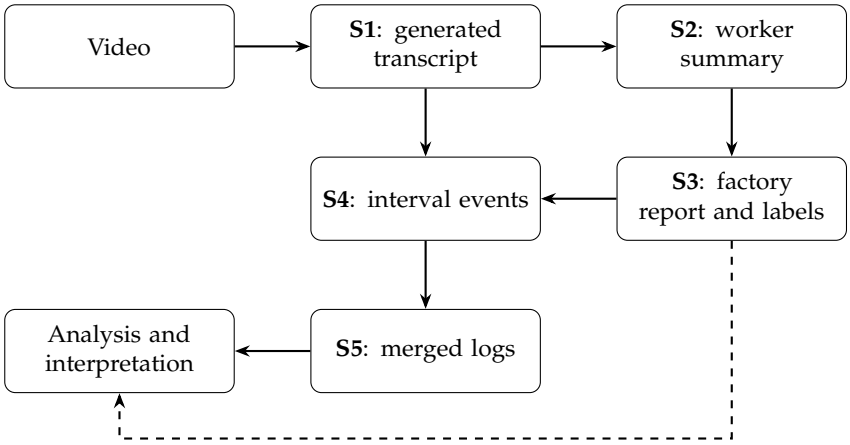


Figure 2. From factory videos to process analysis. The paper explains how we build event logs, presents the data, assesses the logs, and discusses what the results mean. The dashed arrow shows how factory labels guide the analysis categories.

- **A reusable resource.** We link generated transcripts, worker summaries, factory vocabularies, inferred intervals, consolidated logs, and analysis artifacts. This collection supports inspection of the construction steps and reuse of the intermediate text. It is not an independently annotated ground-truth benchmark.
- **Findings about representation and measurement.** Existing examples show how *event boundaries*, *case construction* (choosing which events belong to the same trace), and *category maps* affect duration and sequence descriptions. In factory **003**, transport remains in the event log, but its duration is counted under other categories because the category rule uses broader process labels. We separate these findings from consistency with generated transcripts, visual accuracy, and operational claims that require further evidence.

The assessment combines checks of the released records, archived author judgments against generated transcripts, and descriptive process analyses. Appendix A of the Additional Material summarizes the evidence supporting each type of conclusion.

Four research questions organize the paper. **RQ1:** How are the inferred logs constructed, and which structural properties do the released records satisfy? **RQ2:** How well do event timing and meaning agree with the generated transcripts? **RQ3:** Which descriptive analyses do the logs support, and what limits their operational interpretation? **RQ4:** What general measurement lessons follow from the representation, temporal rules, and shared evidence? Table 1 links each question to its evidence.

2. Background and Related Work

2.1. Unstructured observations as process data

König et al. [7] review how unstructured data enters process mining, including the construction of logs from text, images, and video. Diba et al. [8] distinguish extraction, correlation, and abstraction. These tasks require decisions about which observations become events, how events are grouped, and which concepts describe them. Andrews et al. [9] make data quality part of construction in RDB2Log, which helps analysts derive logs from relational data. Our setting adds a chain of inferred descriptions and labels before a log exists. We examine how choices along that chain affect the resulting measurements.

2.2. Video and first-person workflow understanding

Petersen et al. [5] model and track manual workflows from first-person video to support augmented-reality instructions and feedback during task execution. Pham and

Table 1. Research questions, evidence, and limits.

Question and section	Supporting artifacts	Finding	Main limit
RQ1, Sections 4 and 7.1	Scripts, prompts, reports, all per-video CSVs	Fields, ordering, and vocabulary membership can be checked	Valid structure does not imply correct observation
RQ2, Section 7.2	Generated transcripts, logs, 42 event-support judgments and 42 transcript-entry-overlap judgments	Partial temporal and semantic consistency	No independent video reference; timestamp anomalies
RQ3, Section 7.4	Frozen events, process-category map, reports	Composition, sequence, and worker-profile descriptions	Unequal windows, task mixes, and category meanings
RQ4, Section 8	Worked examples, merge code, metric definitions	Aggregation can hide behavior; missing time has two-sided effects	Conceptual deductions do not estimate unknown errors

Xiao [6] extract assembly steps and operations with semantic descriptions, without pre-labeled training data or pre-trained classifiers. These studies establish first-person workflow understanding as an antecedent. Their goals include following a demonstrated task and recovering its steps.

Our objective is retrospective process analysis across recorded workers. We construct event data to describe time composition, changes between operational categories, and variation in recorded work. A shared vocabulary lets these descriptions use the same names within a factory. It does not establish that workers performed equivalent tasks or that a category transition was a physical interruption. Egocentric video provides access to the wearer's work; the analytical objective and event construction define its role here.

Video process mining also has several established designs. Knoch et al. [10] use hand trajectories and clustering at a camera-equipped assembly workstation. Kratsch et al. [2] develop a reference architecture, and Lepsien et al. [11] combine object tracking, action detection, and abstraction. SnapLog [12] uses pretrained visual representations, temporal segmentation, and a classifier adapted with labeled examples. Its evaluation includes TUM Kitchen and egocentric EPIC-KITCHENS recordings. Thus, camera placement and supervision must be compared for individual methods. The sensor-data review by Brzychczy et al. [13] places these efforts within the broader problem of converting observations into process events.

2.2.1. From video descriptions to process events

Dense video captioning already combines temporal localization with natural-language descriptions [14]. Timestamped narration alone is therefore an established output. Process analysis requires further choices: reusable activity names across recordings, event granularity, selection of operationally relevant episodes, observational case identifiers, source links, and rules for interpreting durations and frequencies.

The video-narration stage's instruction to merge stable repetition illustrates why these choices matter. Several repeated actions can become one described interval and later one event, so event frequency need not equal action frequency. Conversely, the existing foundry example in Section 7.3 turns one 31-second description into two events with shared factory labels and an inferred internal boundary. The added structure enables calculations, while changing what each row and duration represents.

2.3. Video-to-log and language-model-based construction

Gavric et al.'s Vid2log [4] uses LLaVA to infer a domain and relevant activities from selected frames, obtain textual event information, and combine video-derived data with conventional logs through temporal and semantic matching. Both methods use natural language to connect visual observations to event data. Our design first describes recordings, summarizes them by worker, and derives shared factory labels before extracting events

Table 2. Closest construction methods: purpose, input, vocabulary, adaptation, and output. "Not reported" identifies information not established by the inspected source, not an absent capability.

Method	Purpose and input	Vocabulary origin	Task-specific adaptation	Output
Vid2log [4]	Construction and enrichment from video, with conventional logs	Domain and activity prompts	Pretrained LLaVA; domain prompts; parameter updates not reported	Combined event log
Kecht et al. [15]	Construction from customer-service conversations	Supplied topics and activities	Pretrained NLI; labeled threshold calibration and hypothesis selection	XES log
SnapLog [12]	Event discovery from video	Classes supported by labeled examples	Pretrained visual models; classifier adaptation	Events with labels or label distributions
Seeth et al. [19]	Construction from textual reports	Allowed activities from source logs	Pretrained LLMs; supervised fine-tuning compared with prompting	XES traces
This resource	Construction from egocentric factory video	Shared lists induced from worker summaries	Fixed pretrained models; factory-specific context and labels	Interval CSVs and linked text

from the original generated transcripts. This makes the available factory corpus part of the representation used to analyze it. Vid2log studies log richness and an assembly use case; we assess consistency with generated transcripts and the effects of event and category choices on duration and sequence analyses. The comparison therefore concerns the construction of factory context and the assessment of the resulting events.

Kecht et al. [15] use pretrained natural language inference to assign supplied topics and activities to customer-service messages. They use labeled messages to select classification thresholds and refine hypotheses, without training a new language model. Message identifiers and timestamps are retained. Our method estimates episode boundaries from generated video descriptions and derives label lists from the factory material. The comparison therefore involves different sources of event names and times, as well as different uses of human labels.

Shirali et al. [16] apply LLMs to physical observations: they abstract IoT readings into high-level events and integrate logs from multiple sensor sources. LLM use for physical-process data therefore predates this study. Work on process-mining assistants [17] and database-query generation [18] addresses neighboring tasks with different inputs and outputs.

Seeth et al. [19], an arXiv preprint, pair generated reports with source traces, supply permitted activities and attributes, and evaluate extracted traces and process models against source-log references. They report benefits from fine-tuning over zero-shot and few-shot prompting in that setting. Both designs use permitted label lists. Our model parameters remain fixed, while context and shared labels are derived from each factory corpus. We assess consistency with generated descriptions rather than source-log references, so numerical scores from the two studies measure different properties. This comparison does not establish whether fine-tuning would improve our extraction.

Tables 2 and 3 compare the closest transformations using the same method rows. Section 4.3 distinguishes model adaptation from human input in our study.

2.4. Event abstraction and semantic consistency

Event abstraction means choosing the level of detail represented by an event. The review by van Zelst et al. [20] explains how this choice changes process granularity and compares methods by supervision, input, and uncertainty. Here, three questions need separate treatment: where an episode begins and ends, which reusable activity classes exist, and which class is assigned to each episode. Video narration supplies candidate intervals, worker summarization and factory vocabulary construction derive the classes, and event

Table 3. Time, case relationships, and evaluation references for the same methods. A source timestamp, an estimated interval, and an artificial ordering carry different evidence.

Method	Origin of temporal information	Origin of case/entity relationships	Evaluation reference
Vid2log [4]	Frame positions and prompted temporal relations	Temporal/semantic matching and resource-based case inference	Assembly case study and log-richness assessment
Kecht et al. [15]	Recorded message timestamps	Supplied message and conversation identifiers	Labeled messages for classification; process discovery demonstration
SnapLog [12]	Segmented frame sequences	Video sequences as traces	Video datasets with action and boundary annotations
Seeth et al. [19]	Timestamps extracted from generated reports	Report-to-trace pairing from source logs	Source traces and process-level comparisons
This resource	Estimated video-relative intervals; extraction may split spans; merge creates an artificial timeline	Source factory/worker/video metadata; chosen video or worker grouping	Generated transcripts and author audit; computations on inferred logs

Table 4. Neighboring event-data tasks, distinguished by their starting points and intended results. A system can perform more than one task.

Task	Starting point	Intended result
Construction [8]	Observations or documents without a suitable log	An initial event representation
Enrichment [21]	An existing log and additional information	Added or corrected events and attributes
Integration [16,22]	Multiple sources describing related behavior	A combined representation with resolved correspondences

extraction assigns labels and can revise boundaries. Section 4.1 maps these tasks to the implementation.

Inducing classes and assigning instances can fail in different ways. A useful concept can be missing from the vocabulary, or an available concept can be assigned incorrectly. A later category map can also hide a correctly recorded label. The activity and process fields are two label dimensions; the implementation does not establish a parent-child hierarchy. This distinction is necessary to explain factory **003**, where transport activities occur under process contexts that are mapped to other operational categories.

2.5. Construction, enrichment, and integration

Text2EL+ [21] enriches existing logs with events and attributes extracted from text, using semantic and contextual validation and expert guidance. Sturm et al. [22] instantiate parsing, abstraction, extraction, integration, and enrichment using CRM data, email metadata, and call transcripts. Their evaluation involves a sales process and a domain expert. Table 4 separates the tasks these designs can combine.

Our main task is construction. Video narration, summarization, vocabulary construction, and event extraction derive an initial representation. The merge then combines events using source filenames, factory and worker identifiers, and a stated ordering rule (Section 5.4). It does not resolve product correspondences between independently recorded systems. Videos, generated transcripts, and reports are successive representations of the same originating observations. Agreement among them does not provide the independent corroboration that separately recorded enterprise or sensor sources could supply.

2.6. Quality, uncertainty, and relational representations

Suriadi et al. [23] organize recurring event-log imperfections. Our contribution to this concern is an account of how narration, vocabulary construction, assignment, and consolidation can introduce or propagate them. Section 8.2 connects each transformation to an existing check or example and its measurement consequence.

Quality annotations and explicitly uncertain events serve different purposes. Goel et al. [24] retain data-quality information for use during analysis. Pegoraro et al. [25]

represent uncertainty explicitly and compute conformance bounds. SnapLog can retain candidate-label distributions [12]. Our events instead have single-valued activity and process assignments. The audit records particular disagreements, and the five-second boundary tolerance is an evaluation rule. Neither provides calibrated event probabilities or bounds on total downstream error. Structured quality flags and explicit alternatives are possible extensions, with different evidence requirements.

Esser and Fahland [26] represent events and entities across multiple dimensions in graph databases. This complements the case-choice concerns of object-centric process mining [27]. Our worker and video relationships support alternative observational views. Product identities and cross-worker product trajectories require additional evidence; a change in storage format cannot establish them. Similarly, the range of industrial process-mining goals [28] makes it necessary to distinguish descriptions of recorded work from conclusions about production performance.

Building on these methods, we examine how a shared representation derived from factory recordings supports retrospective analysis. The construction and measurement questions focus on how vocabulary, boundaries, case rules, and categories shape the resulting descriptions of work.

3. Problem Definition

The starting point is the factory video in the Egocentric-10K dataset. A camera worn by a worker records visible actions involving tools, materials, and work surfaces. These recordings contain evidence of work, but they do not directly identify the interval events needed for process mining. We therefore first use a vision-language model to produce a generated transcript: timestamped natural-language descriptions of visible actions in each video. We then use large language models (LLMs) to organize the descriptions into shared factory vocabularies and convert the generated transcripts into interval events. Each event should state what action occurred, its broader process context, and its estimated start and end within the recording.

Let F be the set of factories, W_f the worker identifiers associated with factory f, and V_w the recordings associated with worker w. Each video v has an underlying duration D_v, which need not be available in the derived archive. Its generated transcript T_v is an ordered collection of entries (s_i, e_i, x_i), where x_i describes the interval from s_i to e_i. The descriptions and endpoints are model outputs, so the generated transcript is an intermediate representation of the video rather than an independent annotation.

For each factory, the pipeline induces an activity vocabulary A_f and a process vocabulary P_f. Each extracted event records the source recording, worker, factory, estimated start and end times, activity label, and broader process context. The following tuple gives a compact description of one extracted episode in that order:

$$e = (v, w, f, t_s, t_e, a, p), \qquad a \in A_f, \quad p \in P_f. \tag{1}$$

These source links provide provenance, meaning information about where an event and its description came from. The estimated event endpoints t_s, t_e are relative to the video. The generated transcript endpoints delimit descriptions; the extracted event endpoints delimit the events produced from those descriptions. One description can therefore become several events with different boundaries. The LLM interprets T_v using A_f, P_f, and the worker context to produce the event set E_v; the complete log contains $E = \bigcup_v E_v$.

Process analysis also requires a rule for deciding which events belong to the same sequence, or trace. The case function $\kappa : E \to K$ assigns each event to one of these sequences, where K is the set of case identifiers. For example, grouping events by their

Table 5. Notation and intended meaning.

Symbol	Meaning
f, w, v	Factory, worker, and video identifiers; identifiers do not certify roles or task equivalence
D_v, T_v	Source duration and generated transcript; the former is unavailable in the released subset
A_f, P_f	Two separate permitted label lists for factory f
t_s, t_e, d_e	Estimated video-relative start, end, and duration $d_e = t_e - t_s$, in seconds
$\kappa(e)$	Selected case identifier, such as the source video
$g_f(p)$	Analyst's mapping from process label to operational category
C_v, S_v	Interval-union event duration and endpoint span used in coverage diagnostics

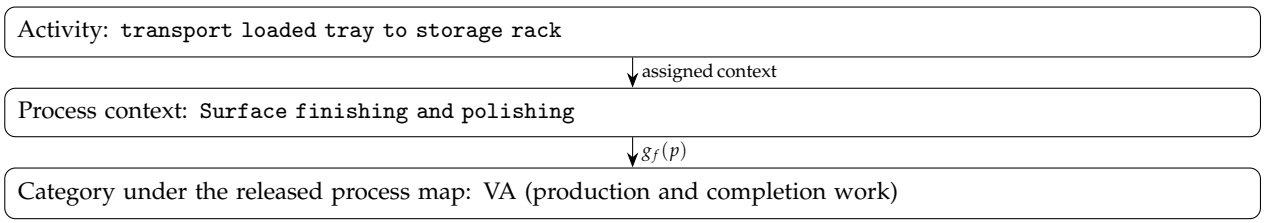


Figure 3. A transport action can contribute to value-adding (VA) time under the authors' process-based category rule for production and completion work. Source: factory **003**, worker 001, recording 00001, event row 2, from 00:03:52 to 00:04:00 (8 seconds). The action is recorded; its duration enters the VA category.

source video makes each recording a case. Section 5.4 compares the available case choices. The construction problem is thus to transform videos into generated transcripts and then into labeled intervals, while retaining enough source information to inspect the resulting log. Vocabulary induction uses descriptions across the available factory recordings, so event extraction depends on both the local generated transcript and its factory context.

Table 5 summarizes the notation. An event represents an operational episode. One episode may summarize many repeated actions, and several events may divide a single narrated interval. It need not be one item, one production cycle, or one complete process instance. Desirable properties include faithful descriptions, meaningful boundaries, appropriate labels, and sufficient coverage. The implementation enforces only a subset: parseable offsets, end not before start, required values, and membership in the separate vocabularies. Section 4.3 distinguishes these checks from intended properties.

3.1. Activity, process context, and analytical category

An action, its broader production context, and the category used to summarize its duration are different descriptions of the same event. An *activity* answers what action is performed. A *process context* identifies the broader operation to which the extractor assigns it. An *operational category* describes how an analyst groups the action or context for a particular question. The implemented checks verify the two labels separately: $a \in A_f$ and $p \in P_f$. A compatibility relation $H_f \subseteq A_f \times P_f$ would specify which activity and process pairs are allowed together. The implementation enforces neither this relation nor a process-to-activity hierarchy.

For the reported category analysis, the assigned process label determines the operational category through $g_f(p)$. VA denotes value-adding work under the authors' operational scheme for production and completion work; it does not directly measure customer value. Non-VA refers to all categories outside VA. Thus, transport within surface finishing can receive the VA category assigned to finishing, as Figure 3 shows.

Figure 3 gives a released example of the process-based rule $g_f(p)$. An explicit compatibility relation or a category rule using (a, p) could improve future analyses, but neither

Table 6. Conceptual tasks and their roles in the five-stage construction. The stages interact; this is not a claim that each task has an isolated error rate.

Task	Main stage	Decision made
Temporal segmentation	**S1**	Describe candidate episodes with estimated boundaries; merge stable repetition
Activity-class construction	**S2** and **S3**	Aggregate worker descriptions and induce reusable factory activity and process lists
Event-instance assignment	**S4**	Select permitted labels for episodes; retain or split generated transcript intervals and estimate internal boundaries

Table 7. Stage specification from scripts 01 to 05. A nonempty text output is not a validation of its factual content.

Stage; unit	Inputs and context	Output	Failure and reuse behavior
S1; video	Uploaded video; transcription prompt	One chronological text file	Keeps existing files; retries transient failures; rejects an empty response; attempts deletion of uploads
S2; worker	All saved generated transcripts in one request; filenames omitted from the bundle	Worker report	Missing or empty input raises an error; empty response rejected; keeps existing output unless overwrite is set
S3; factory	All worker summaries in one request, with worker identifiers	Factory report with two label lists and compact worker entries	Same text and reuse checks as **S2**; malformed section structure may fail later in **S4**
S4; generated transcript	Generated transcript; two label lists; compact **S3** worker entry, or a missing-context marker	Per-video CSV	Missing lists or invalid events raise errors; an empty event list is valid; existing CSV kept unless overwrite is set
S5; worker, then factory	CSVs in sorted filename order	Worker and factory CSVs	Checks timestamp columns and consistent headers; empty files add no time; keeps existing output if its schema matches

was enforced in this extraction. An activity-based category rule would define a different measurement of the recorded work.

4. Pipeline and Design Rationale

The released scripts implement five stages. **S1** describes the visible actions in each video as a generated transcript. **S2** summarizes each worker's generated transcripts. **S3** combines the worker summaries into a factory report and shared label lists. **S4** extracts events from each original generated transcript using those lists and the compact worker entry from the **S3** report. **S5** merges the resulting CSVs. Figure 2 in the Introduction gives an overview of these stages. Reports and logs share evidence, so their agreement is not independent validation.

Adaptation takes place in the context and vocabulary, while the foundation-model parameters remain fixed. Each available factory corpus contributes to its own label lists through **S2** and **S3**, and **S4** then uses those lists to label recordings from that corpus. The six-factory assessment therefore applies the same construction procedure to six separately induced representations. It is a retrospective use of the available material, not an evaluation of a fixed vocabulary on a previously unseen factory. Prompts also supply analyst choices about operational relevance and granularity.

4.1. Vocabulary construction as event abstraction

Event abstraction concerns the level at which observations represent process behavior [20]. Table 6 separates three decisions in our implementation. Constructing a reusable class and assigning an event to it are different tasks: a missing concept and a mistaken assignment require different checks. The process list supplies a second label dimension, with no enforced parent-child relation to the activity list.

```
for each selected factory f:
    for each available video v:
        T[v] = reuse existing transcript or S1(video[v], prompt1)
    for each worker w:
        B = concatenate all T[v] for w, without filenames
        R[w] = reuse existing summary or S2(B, prompt2)
    F[f] = reuse existing report or S3(all R[w] for f, prompt3)
    A, P, worker_context = parse the required sections of F[f]
    for each transcript T[v]:
        context = T[v] + A + P + worker_context[w]
        E[v] = reuse CSV or normalize(S4(context, prompt4))
    for each worker w:
        offset = 0
        for each CSV v in sorted filename order:
            append E[v] shifted by offset; keep local offsets
            if E[v] is nonempty: offset += max end(E[v])
    concatenate worker logs into the factory log
```

Listing 1: Algorithm 1: implemented dependency order. API retries surround requests; this pseudocode does not add windowing or new validation.

4.2. Execution units and failure behavior

Table 7 states the input, output, and failure behavior for each execution unit. The algorithm below gives their dependency order. Listings 2–5 reproduce the four full prompt templates, with punctuation normalized for display. The original templates and their hashes remain in the resource. Placeholder braces are filled by the scripts; the additional context is specified in Table 7. The prompts specify requested model behavior, not guaranteed output properties.

The **S1** prompt (Listing 2) asks for a new timestamped entry when an action, object, location, or interaction changes. It asks for repeated stable work to be summarized once and for uncertainty to be stated briefly. These are instructions to the model. **S1** does not check whether each visible change was narrated or whether the timestamps match the video.

The **S2** worker-summary prompt (Listing 3) and the **S3** factory-report prompt (Listing 4) compress large text bundles into summaries. This makes factory-wide vocabulary induction possible, but uncommon actions can be lost. The implementation does not automatically split long requests into windows, reconcile separate summaries, or check coverage against all generated transcript spans. **S4** partly bypasses this compression by reading each original generated transcript, although it must still use the induced vocabularies.

The **S4** event-extraction prompt (Listing 5) requests a strict JSON schema with estimated start and end times, activity, and process. Its local parser can also accept a list, strip code fences, recognize fallback time keys, and parse two-component offsets or numeric seconds. It normalizes label case and surrounding quotes, checks membership, rejects end-before-start intervals, and sorts rows by start, end, activity, and process. It allows zero-duration events and overlaps. It does not check video bounds, full temporal coverage, the truth of an activity, or compatibility between the activity and process. The archive contains CSV outputs, not the original **S4** JSON responses.

S1 to **S3** default to `gemini-3.1-pro-preview`; **S4** defaults to `gpt-5.4`, requests high reasoning effort, and permits up to 100 concurrent extraction tasks. These are recoverable code settings. Per-request execution records are unavailable. No temperature, token limit, seed, cost, or runtime is inferred from missing records. Appendix B of the Additional Material specifies the distinction between code defaults and historical execution evidence.

4.3. Requests, guarantees, observations, and expectations

Table 8 distinguishes requested, enforced, observed, and untested properties. "Automated extraction" means that generation uses no task-specific fine-tuning or labeled event

```
Please transcribe the attached egocentric video into a very detailed chronological text description.

I do not want a formal event log, structured schema, activity labels, or any kind of post-processing. I only want plain text entries, each
beginning with a timestamp range, followed by a detailed description of what happens in that time span.

Requirements:
- Format every entry as:
  [HH:MM:SS - HH:MM:SS] description
- Follow the video strictly from start to end.
- Use short timestamp intervals whenever something changes.
- Be highly detailed, but do not repeat the same type of behavior again and again when nothing changes.
- If the same action continues in the same way, with the same context and no relevant difference, merge it into one longer timestamped entry
 instead of splitting it into many repetitive entries.
- Only create a new entry when something meaningful changes, such as:
  - the action changes
  - a new object is handled
  - the person changes location
  - attention shifts to something else
  - another person becomes involved
  - something unusual, problematic, interrupted, or different happens
- Capture micro-actions when visible, such as reaching, grasping, lifting, placing, opening, closing, turning, walking, pausing, looking,
adjusting, and interacting with objects or people.
- Describe what is visible in the egocentric view, including hand movements, object handling, scene changes, and shifts of attention.
- If a repeated behavior continues in exactly the same way, summarize that repeated stretch once with an appropriate timestamp range instead
 of listing each repetition separately.
- If the scene is partially unclear, mention the uncertainty briefly instead of guessing.
- Do not use tables, bullets, JSON, labels, categories, or analysis language.
- Do not convert the result into an event log.
- Do not add any summary at the end.
- Only provide the timestamped descriptive transcription.

The output should read like a detailed time-aligned narration of the video, not like a structured annotation file.

Example of the intended compression:
- Good:
  [00:03:10 - 00:03:42] The wearer continues wiping the same section of the table with repeated back and forth hand movements. No new object
 or interruption appears.
- Not desired:
  [00:03:10 - 00:03:14] The wearer wipes the table.
  [00:03:14 - 00:03:18] The wearer continues wiping the table.
  [00:03:18 - 00:03:22] The wearer continues wiping the table.

Please produce the full transcription in this style for the entire video.
```

Listing 2. Full video transcription prompt for **S1**.

Table 8. Four levels of evidence for pipeline properties.

Level	Example	What can be concluded
Prompt instruction	Mark uncertainty; include handling and waiting when visible; split meaningful changes	The behavior is requested. Compliance and completeness require separate assessment.
Software constraint	Parse event fields; canonicalize permitted labels; reject negative intervals; sort rows	Accepted outputs satisfy these checks. Semantic correctness and accurate time remain open.
Observed artifact property	All 10,698 rows have required values and permitted labels; 35 adjacent overlaps are detected	These statements hold for this release. They are not promises about future runs.
Untested design expectation	Summaries help vocabulary consistency; closed labels help interpretation	These are rationales. No ablation or comparative benefit estimate is available.

Table 9. Supervision and human input in this study. Unlike labeled threshold calibration in Kecht et al. [15] or parameter adaptation in SnapLog and Seeth et al. [12,19], generation here adapts through prompts and induced context.

Dimension	Role in the present procedure
Model pretraining	General-purpose pretrained models provide visual and language capabilities
Task-specific parameter updates	None in **S1** to **S4**; no model is trained or fine-tuned on these factory recordings
Labeled examples	No labeled event examples for training, prompting, or calibration; **S1** includes an illustrative narration-format example
Vocabulary origin	**S3** induces two lists from the available worker summaries; **S4** checks list membership
Domain and analyst input	Authored prompts define relevance and segmentation; analysts choose case projections and the operational-category map
Human assessment	One author's archived event and transcript-entry judgments assess outputs; they are not generation inputs

examples. It does not mean that the complete study is free of human judgment. Table 9

```
You are given multiple detailed chronological transcriptions from egocentric factory videos recorded by the same worker.

Your task is to synthesize those transcriptions into a process-focused report about what work is being done and what process-level
optimization opportunities are visible.

Context:
- Factory identifier: {factory}
- Worker identifier: {worker}

Important output constraints:
- Do not mention the factory identifier, worker identifier, file names, file paths, transcript numbers, video numbers, or any other source
identifiers.
- Do not mention timestamps unless they are absolutely necessary, which they normally are not.
- Do not quote the transcripts.
- Do not write a per-video recap. Synthesize across all provided transcriptions.

Required focus:
- Identify the process or processes performed by the worker.
- If the worker appears to perform more than one distinct process, separate them clearly.
- Describe the operational flow in enough detail that someone can understand how the work is carried out.
- Include relevant details such as setup, repeated action loops, machine interaction, material handling, part movement, quality checks,
cleaning, waiting, transport, handoffs, and interaction with nearby stations when visible.
- Stay grounded in the provided evidence. If something is uncertain, state the uncertainty briefly instead of inventing details.

Optimization constraints:
- Only suggest process optimizations that are visible or reasonably inferable from the observed workflow.
- Focus on process design, workstation layout, batching, staging, transport, sequencing, buffering, machine idle time, tool/material
presentation, inspection placement, handoffs, or coordination between steps.
- Do not suggest worker-level optimizations such as "work faster", "be more careful", "improve discipline", "train the worker better", or
similar people-focused advice.
- If no credible process optimization is evident, say so.
- When an optimization is inferential rather than directly observed, label it clearly as an inference.

Write the output in plain text with exactly these section headings:

Process Summary
Detailed Process Breakdown
Possible Process Optimizations
Evidence Limits
```

Listing 3. Full worker-summary prompt for **S2**.

Table 10. Design rationale and conceptual alternatives. No alternative in the last column was experimentally compared here.

Choice	Intended benefit	Limitation	Possible alternative
Video-to-text intermediate	Readable evidence; reuse for different analyses	Visual errors become shared textual evidence	Direct video-to-events, with stronger visual provenance
Worker summaries	Compact factory-level context	Rare behavior and recording identity may be lost	Window summaries with explicit source references
Factory vocabularies	Concrete local labels across workers	Different factories use different abstractions	Global concepts with local extensions
Closed labels	Prevent out-of-list output	Closest allowed label may still be wrong	Open labels followed by review and reconciliation
Change-point narration	Compact description of stable behavior	Boundary detail and repeated cycles can disappear	Finer segments with explicit repetition counts
Interval events	Preserve approximate episode duration	Row counts do not equal production volume	Object-linked cycle records, when supported

identifies the different inputs. “Constrained” refers to the separate permitted label lists. “Validated” is used only with a named check, never as a general claim of visual accuracy.

4.4. Design choices and tradeoffs

Table 10 summarizes the design choices and alternatives. Repeated cycles deserve particular care. If ten cycles become one narrated interval and one event, the event count decreases and the directly-follows relation loses nine possible repetitions. A longer event can represent more repetitions, slower work, or more aggressive compression. The logs alone do not distinguish these explanations. The text interface makes this decision inspectable, but does not recover details omitted from the generated transcript.

```
You are given multiple worker-level process summaries from the same factory.

Your task has two goals:
1. Produce a factory-level report that restates the worker process summaries in a compact but informative way, preserving which worker each
summary belongs to.
2. Produce a long, detailed, process-mining-oriented activity catalogue for the factory that can help with downstream process optimization.

Context:
- Factory identifier: {factory}

Important constraints:
- Do not mention file names, file paths, source identifiers, transcript numbers, or video numbers.
- Do not quote the source summaries.
- Stay grounded in the provided evidence. If something is uncertain, state the uncertainty briefly instead of inventing details.
- Synthesize across workers. Do not merely copy the worker summaries back verbatim.

Worker summary requirements:
- Include every worker provided in the input.
- Keep each worker summary concise but useful.
- Preserve the worker identifier for each summary, because the output is factory-level.
- Focus on the main process(es), major material flows, handoffs, machine interactions, transport, inspection, waiting, cleaning, batching,
and coordination patterns that matter operationally.

Factory process label requirements:
- Explicitly provide one raw deduplicated list of processes executed across the different workers in the factory.
- Express each process as a short label only.
- Keep process labels concise and comparable across the factory.
- Prefer labels such as "metal stamping", "box assembly", "garment bagging", "part transport", "inspection", or "machine cleaning".
- Distinguish clearly between materially different processes, but do not create unnecessary micro-variants.
- Do not group the labels by worker.
- Do not repeat the same label for multiple workers.

Activity catalogue requirements:
- The catalogue must be long and detailed.
- Each activity must be a short string only, suitable as a candidate event label for process mining.
- Prefer specific verb-object phrases such as "load raw blank into press" rather than generic labels such as "operate machine".
- Include not only value-adding operations, but also transport, staging, waiting, inspection, rework-like handling, setup, cleanup, handoff,
 documentation, and coordination activities when visible.
- Keep distinct activity variants separate when they matter for process analysis.
- Avoid duplicate or near-duplicate labels.
- Avoid full sentences, explanations, numbering inside the activity text, or extra commentary next to the activity.
- If an activity is not clearly supported, do not invent it.

Optimization relevance requirements:
- Favor activity labels that would help reveal bottlenecks, queues, machine idle time, material handling burden, unnecessary motion,
batching behavior, rework, inspection points, coordination dependencies, and handoff structure.

Write the output in plain text with exactly these section headings:

Factory Process Overview
Worker Process Summaries
Factory Process Labels
Factory Activity Catalogue
Optimization-Relevant Process Observations
Evidence Limits

Formatting rules for the output:
- Under "Worker Process Summaries", use one line per worker in the form:
  worker_xxx: summary
- Under "Factory Process Labels", use one bullet per process label.
- Each bullet in "Factory Process Labels" must contain only a short process label and nothing else.
- Under "Factory Activity Catalogue", use one bullet per activity.
- Each bullet in "Factory Activity Catalogue" must contain only a short activity string and nothing else.
```

Listing 4. Full factory-report and vocabulary prompt for **S3**.

5. Dataset and Released Artifacts

5.1. An inferred resource for process analysis

This resource links generated transcripts, worker summaries, factory vocabularies, inferred events, logs, and analysis artifacts from manual work in six factories. Readers can inspect how the pipeline constructs events and how those events support process measurements. The released labels and boundaries are inferred by the pipeline.

Assembly101 [29], by comparison, records toy assembly and disassembly with simultaneous static and egocentric cameras. Its coarse and fine action annotations support benchmarks for recognition, anticipation, and temporal segmentation. The present resource

```
You are converting one timestamped raw factory annotation into a structured event list for process mining.

Context:
- Factory identifier: {factory}
- Worker identifier: {worker}

You will be given:
- the allowed process labels for the factory
- the allowed activity labels for the factory
- an optional worker summary for additional context
- one raw annotation transcript with timestamp ranges measured from the start of the video

Your job:
- Convert the raw annotation into a JSON object with one key, ‘events‘, whose value is a list of event objects.
- Each event must correspond to one operationally meaningful event segment.
- Use the transcript timestamps as the main basis for the event boundaries.
- If a transcript interval contains one repeated stable activity, keep it as one longer event.
- If a transcript interval clearly contains multiple different events, you may split it into multiple estimated events.
- Keep the events in chronological order.

Strict label constraints:
- The value of ‘activity‘ must be exactly one label from the allowed factory activity list.
- The value of ‘process‘ must be exactly one label from the allowed factory process list.
- Do not invent new activity labels.
- Do not invent new process labels.
- If multiple labels are plausible, choose the closest allowed label.
- Be especially careful to respect these two controlled vocabularies exactly.

Output schema:
- Return JSON only.
- Return a top-level JSON object with exactly one key: ‘events‘.
- The value of ‘events‘ must be a JSON array.
- Each array item must be an object with exactly these keys:
  - ‘estimated_start_time‘
  - ‘estimated_end_time‘
  - ‘activity‘
  - ‘process‘
- ‘estimated_start_time‘ and ‘estimated_end_time‘ must be strings in ‘HH:MM:SS‘ format, measured from the start of the video.

Additional rules:
- Include only process-relevant operational events. Ignore purely accidental camera motion or camera removal unless it directly reflects a
process-relevant activity.
- Do not output markdown fences.
- Do not output explanations, notes, comments, or any text before or after the JSON object.
- If no valid operational events can be extracted, return ‘{"events":[]}‘.
```

Listing 5. Full event-extraction prompt for **S4**.

serves a different purpose: its linked outputs make the construction and interpretation of process measurements inspectable.

Researchers can use the preserved generated transcripts to study alternative event representations, inspect which descriptions support a label, or compare how activity and process categories affect an aggregate. Such comparisons would examine representation choices unless independently checked against video. The present release includes structural checks and an author audit against generated transcripts, but no independent video annotations. Its inferred labels are not an independently annotated ground-truth benchmark for action recognition or boundary detection.

5.2. Selection and provenance

The archived subset contains outputs for factories **001** to **006**, with 59 worker identifiers and 322 paired generated transcript and event files. Factory and worker identifiers are inherited from the source organization. Domain descriptions in this paper summarize generated factory reports; they are not independently verified factory metadata. The archive does not record the selection rationale, the original source revision, or a complete inventory of the recordings available at extraction time. We therefore describe an available processed subset, not a random or representative sample of factories.

The current provider card reports a median clip length of 180 seconds [3], whereas many archived generated transcript endpoints lie near 20 minutes. These describe different quantities and possibly different snapshots or subsets. The card documents a per-video

Table 11. Artifact inventory. Counts come from the frozen repository outputs. Worker and factory merged logs repeat the same source events.

Layer	Files	Content and provenance
Generated transcripts	322	One text per video; recording identity in path
Worker summaries	59	One text per worker; combines generated transcripts
Factory reports	6	Narrative, worker entries, activity and process lists
Per-video logs	322	10,698 interval-event rows; six stored columns
Merged worker logs	59	Same 10,698 events with source file, worker case, and local offsets
Merged factory logs	6	Same 10,698 events concatenated across workers
Audit record	1	42 event-support judgments and 42 transcript-entry-overlap judgments
Prompts; stage scripts	4; 5	Extraction instructions and implementation

`duration_sec` field inside source archives, but those metadata files are not included here. The pipeline scans video files and does not preserve that field. The selected recordings' actual durations therefore cannot be established from the derived archive. The inspected public dataset snapshot does not identify the historical source version used for extraction.

The six per-video columns are `start_timestamp`, `end_timestamp`, `activity`, `process`, `factory`, and `worker`. **S5** adds `source_file`, `case_id`, video-relative start and end offsets, and duration. Table 11 and the hashed artifact manifest distinguish file counts from unique event counts. The released dataset documentation follows the documentation concerns of Datasheets for Datasets [30]: composition, collection, uses, and limitations.

There are 13,304 generated transcript entries recognized by the original three-component timestamp parser. Recognizing the additional two-component and mixed-format entries gives 13,462 entries, without rescaling any timestamp.

Across all logs, the original endpoint-span calculation totals 786,412 seconds (218.4 hours). This sums parsed spans, not verified footage durations. Interval unions cover 414,886 seconds by counting overlapping time only once. Summed event durations total 421,283 seconds because each row contributes its full duration. Their difference, 6,397 seconds, is the additional time counted through overlaps.

The retained time-analysis subset contains 10,647 events in 320 logs, with 349,659 seconds of summed event duration (97.1 hours). This is the duration used for category shares; it is neither the full archive's endpoint span nor its interval-union duration.

5.3. Temporal data quality and inclusion rules

Unusual generated transcript timestamps and exclusion from the time analysis are separate issues. The inclusion rule is applied to extracted event endpoints: a log is excluded when its largest event endpoint exceeds one hour. The original parser recognizes three-component timestamps; the broader parsing diagnostic also recognizes two-component and mixed-format entries. This explains the different entry counts above.

Table 12 identifies all seven generated transcripts whose literal parsed endpoint exceeds one hour. An endpoint of `20:06:00` paired with an event endpoint of `00:20:06` is consistent with a format error, but does not prove which timestamp is correct. Some files switch from `HH:MM:SS` to `MM:SS`. The analysis retains the recorded timestamps.

The original value-analysis code excludes a log only when its maximum event endpoint exceeds 3,600 seconds. It excludes two logs, both in factory **001**. This rule retains five logs associated with generated transcripts with unusual timestamps. Retention is a calculation rule, not certification of their timestamps.

Four other generated transcripts contain two-component endpoints. Their identifiers are **001**/004/00001, **002**/005/00001, **002**/009/00003, and **003**/003/00003. Three change format within the file; **002**/005/00001 uses `MM:SS` throughout. The original parser misses

Table 12. Generated transcript and event endpoints for the seven generated transcripts with literal spans over one hour. All displayed timestamp strings are read as hours:minutes:seconds. The final column applies the original time-analysis rule to event endpoints. IDs abbreviate factory/worker/video. Source durations are unavailable for all rows.

ID	Generated transcript endpoint	Event endpoint	Time-analysis decision
001/001/00004	20:00:00	20:00:00	Exclude
001/010/00001	01:20:00	01:20:00	Exclude
002/002/00008	20:00:00	00:20:00	Retain
002/010/00001	20:00:00	00:20:00	Retain
005/002/00000	13:52:00	00:13:52	Retain
006/001/00001	20:06:00	00:20:06	Retain
006/004/00000	19:59:00	00:19:59	Retain

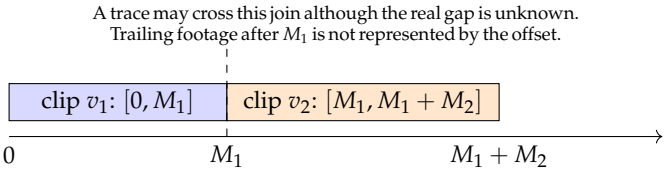


Figure 4. Schematic of the implemented merge, not measured source time. The first clip occupies the synthetic span from zero to its last extracted endpoint; the second starts there and adds its own endpoint span. Separate video cases preserve the clip boundary. A worker case creates a directly-follows relation across it.

158 entries across these files. The inclusion manifest records every video's formats, format-transition positions, endpoints, row count, union duration, summed duration, and decision. A separate interval manifest preserves the original endpoint strings. The low coverage ratios in Section 7.2 therefore diagnose inconsistent representations; they do not establish long-video truncation.

5.4. Synthetic time and case selection

Each recording has its own local timeline. To combine recordings into a worker sequence, the merge shifts their event times onto a common artificial timeline. **S4** stores local offsets as dates relative to 2000-01-01 for software compatibility. The archive provides no verified calendar times or cross-worker synchronization. **S5** processes clips in filename order, which is not proof of their real chronological order.

For a worker's ordered clips $v_1, \ldots, v_m$, the last extracted event endpoint in recording j is $M_j = \max_{e \in E_{v_j}} t_e$, with $M_j = 0$ for an empty log. Let O_j be the accumulated time shift for that recording, and let the primed endpoints be the shifted event times. **S5** adds the preceding recording's endpoint to the accumulated shift and applies it to each event:

$$O_1 = 0, \qquad O_{j+1} = O_j + M_j, \qquad (t'_s, t'_e) = (O_j + t_s, O_j + t_e). \tag{2}$$

The same shift is added to both endpoints, preserving the event's duration. For example, if the first recording's last extracted endpoint is 60 seconds, an event at 10 to 20 seconds in the next recording appears at 70 to 80 seconds. The next clip starts at the largest extracted endpoint, not at the full source duration. Unrepresented trailing footage and empty logs add no elapsed time. Within-clip gaps before that endpoint remain on the synthetic axis, but gaps between recording sessions are unknown. Each worker has a separate synthetic origin; equal synthetic dates do not establish simultaneous work.

Figure 4 illustrates the synthetic clip join, and Table 13 compares the case rules supported by the released attributes.

Table 13. Case constructions supported by the attributes. Process episodes require an additional grouping step and are not stored as case IDs.

Case rule	Supported description	Interpretation risk
Video as case, $\kappa(e) = v$	Local activity order within an observation window; used for discovery	Window may start or end midway through a production process
Worker as case, $\kappa(e) = (f, w)$	Sequence of retained clips; stored **S5** case ID	Filename joins can create artificial transitions; calendar duration is unknown
Contiguous process episode	Consecutive events with the same process within a chosen clip, with a stated gap rule	Inferred context boundaries do not identify items or completed instances

Table 14. Evaluation questions, required references, and evidence in this study. Successful computation and correct operational interpretation are separate claims.

Question	Reference needed	Present evidence and scope
Are visual actions correct?	Independent annotations of the video	Not measured; generated transcripts cannot detect their own visual errors
Are labels consistent with text?	A reference interpretation of the text	Existing author judgments for 42 events; label membership checked for all rows
Are boundaries accurate?	Independently defined action boundaries	Agreement with generated transcript endpoints only; the five-second tolerance is a matching rule
Can the log support process-mining computations?	Defined case rules, measures, and completed calculations	Discovery and descriptive outputs demonstrate computation on the inferred representation
Are operational conclusions correct?	Valid measurements and evidence relevant to the conclusion	Traceable category descriptions and hypotheses; no independent production outcomes or intervention evidence

6. Assessment Methodology

The assessment examines the structure of the extracted events, their consistency with generated descriptions, and their use in the reported descriptive analyses. It combines structural checks, the existing author audit, and calculations with explicitly defined measures.

The inputs are frozen texts, reports, CSVs, and archived author judgments. The descriptive workflow runs offline with Python's standard library. Activity discovery uses PM4Py [31] with the archived miner and noise-threshold settings. All calculations use the released records as their input.

6.1. Evaluation questions and reference data

Table 14 distinguishes the objects of evaluation. Assembly101 provides video action annotations [29]; Seeth et al. compare text-derived logs with source traces and process-level references [19]. Our generated transcript checks assess an intermediate transformation, while our downstream calculations demonstrate uses of its output. These reference structures support different conclusions, so their accuracy, fitness, and precision values should not be ranked as if they measured the same property.

6.2. Metrics, denominators, and aggregation

The measures describe different properties of the inferred logs. We therefore specify what each measure counts, which observations enter its denominator, and how results are combined across recordings or workers.

The coverage diagnostic measures how much of the parsed timestamp span is occupied by at least one extracted event. For each video, let E_v be its events and B_v the set of generated transcript endpoints recognized by the original parser. Let $T_v^{\max}$ be the maximum such

endpoint and M_v the maximum event endpoint. We first calculate occupied time and the endpoint span for each recording, then divide their factory totals:

$$C_v = \left| \bigcup_{e \in E_v} [t_s, t_e) \right|, \qquad S_v = \max(T_v^{\max}, M_v), \qquad \text{coverage}_f = \frac{\sum_{v \in f} C_v}{\sum_{v \in f} S_v}. \tag{3}$$

C_v measures the seconds occupied by events, counting overlapping intervals only once. The vertical bars around the union of intervals mean its duration in seconds. S_v measures the span from zero to the latest recognized generated transcript or event endpoint. The factory diagnostic divides total occupied seconds by total endpoint span. Recordings with longer endpoint spans contribute more to the result, so the ratio is weighted by span rather than an average of recording percentages. Its denominator comes from recorded endpoints, not verified footage durations. When no generated transcript endpoint is recognized, the event endpoint supplies the denominator. Results from the broader parser are released as a separate diagnostic; the original table is retained and labeled.

Boundary agreement measures how often an event's start or end lies within five seconds of a recognized generated transcript endpoint. This five-second tolerance is the matching rule in Table 15. Each event contributes two boundaries to the denominator. If no generated transcript endpoint is recognized for a recording, its event boundaries count as unmatched and remain in the denominator. Several event boundaries may match the same generated transcript endpoint. A match need not identify the correct action or justify an internal split.

Category shares describe how the retained event duration is distributed across the categories in the chosen process map. Let E_f^* contain the factory's events retained for time analysis under $M_v \leq 3600$, and let E_w^* be the retained events for worker w. For each category, we sum the durations of events assigned to it and divide by the summed duration of all retained events. The process label determines the category. Each event row contributes its duration, including time that overlaps another row. Coverage instead counts that shared time only once.

Worker summaries use the same retained events. Worker event-hours H_w come from summed event durations, converted from seconds to hours. We calculate each eligible worker's VA share first and then take the factory median, giving workers equal weight. For grouping, the dominant process is the one with the largest summed duration; subsequent worker statistics use all of that worker's retained events. In Table 15, # counts the indicated events or list elements, the indicator **1** equals one when its condition is satisfied and zero otherwise, and arg max selects the process with the greatest summed duration.

Table 15 defines the measures and their denominators. For category analysis, the process lists have 113 factory-specific entries and 108 distinct strings across factories. Of the 113 entries, 112 are used in the logs. The unused factory **005** label *motor assembly finishing* has no category assignment in the original map. Every process string actually assigned to an event is mapped. Appendix D of the Additional Material lists the map by factory and marks the unused entry explicitly.

The categories are value-adding production work (VA), material handling and staging (MH), transport (TR), quality inspection (QI), cleaning and maintenance (CM), rework (RW), and documentation (DOC). VA includes packaging and other completion work under the authors' scheme; it is not a direct measurement of customer value. Defect patching is assigned RW because its label and descriptions concern filling or repairing surface defects. This does not establish that it is avoidable or that the upstream process failed a specified quality standard. Supporting work can be necessary even when assigned outside VA.

Table 15. Metric definitions. E_f^* is the retained set under $M_v \leq 3600$; E_w^* is its worker subset. Summed event-hours are $H_w = \sum_{e \in E_w^*} d_e/3600$. R_w^{VA} is the worker's list of VA run durations, converted to minutes. Boundaries in recordings with no recognized generated transcript endpoints count as unmatched.

Quantity	Definition	Unit, aggregation, exclusions
Vocabulary membership	$\#\{e : a \in A_f,\ p \in P_f\}/\#E_f$	Fraction of all event rows
Vocabulary use	Distinct assigned labels divided by permitted labels	Separate activity and process lists, per factory
Coverage diagnostic	$\sum C_v / \sum S_v$	Fraction of parsed endpoint span; all logs
Boundary agreement	$\sum_{e,b \in \{t_s,t_e\}} \mathbf{1}[\min_{t \in B_v} \lvert b - t \rvert \leq 5]/(2\#E)$	Fraction of boundaries; all logs; original parser
Category share	$q_{fc} = \sum_{e \in E_f^*} d_e \mathbf{1}[g_f(p_e) = c] / \sum_{e \in E_f^*} d_e$	Fraction of summed event duration; overlap counted in each row
Worker VA share	Same ratio restricted to E_w^* and category VA	Eligible workers require $H_w \geq 0.5$; factory median weights workers equally
VA transition rate	I_w/H_w, where I_w counts VA-to-non-VA transitions in concatenated run lists	Transitions per summed event-hour; factory median
Mean VA run	$\overline{R}_w = \sum R_w^{VA}/\#R_w^{VA}$	Minutes; table reports median across eligible workers of these means
MH/TR count and rate	Number of retained MH or TR rows; count divided by summed event-hours	Rows, not merged episodes; duration-weighted factory denominator
Dominant process	$\arg\max_p \sum_{e \in E_w^*: p_e = p} d_e$	Groups workers; comparisons then use all their retained events
Self-loop share	Equal adjacent activity labels divided by all adjacent pairs	Within video cases only; all logs

Consecutive events with the same category form a run within each recording. A run's duration is the sum of its event durations, which can bridge unrepresented gaps and count overlaps. We divide run durations in seconds by 60 to express them in minutes. We calculate the mean VA-run duration for each eligible worker and then report the median of those worker means. No eligible worker has an empty VA-run list in this release; the original implementation would report zero for that case.

The worker's run sequences are concatenated, and changes from VA to another category are counted and divided by the worker's summed event-hours. This can count transitions across a clip join; the release contains 15 such transitions among eligible workers. The rates depend on the event representation and do not verify interruptions of continuous work. The original script rounds each worker's VA share, transition rate, and mean VA-run duration to one decimal before taking factory medians. We retain this rounding rule.

All per-video logs are nonempty. Negative and zero durations are checked separately; no such rows occur. Overlaps remain in time composition and are merged only for coverage. Ambiguous generated transcript timestamps remain literal. This preserves the original calculations while making their limits visible.

6.3. Existing semantic audit protocol

The audit asks two questions. The event sample checks whether labels are supported by overlapping generated text. The generated transcript-entry sample checks whether sampled entries overlap extracted events. The archive records 42 event-support judgments and 42 transcript-entry-overlap judgments, seven of each per factory. One author made the judgments. The audit record documents a random seed of 42 and factory stratification, but the complete sampling implementation is unavailable. The sampling order, full sampling frame, and handling of timestamp formats cannot be reconstructed. We report the existing sample as documented and do not claim that it can be redrawn exactly.

For the event sample, the reference is the generated text overlapping the event interval. Code C means fully supported: the label describes the action in that interval's generated transcript. P means partially supported: the label is plausible but the overlapping text alone is insufficient. X means contradicted. Of the 42 sampled events, 36 are fully supported and six are partially supported. Dividing the fully supported count by all sampled events gives

Table 16. Corpus and generated transcript diagnostics using the original parser. Activity and process columns show permitted/used labels. Coverage divides occupied event seconds, counting overlaps once, by parsed endpoint spans rather than verified video duration. Boundary agreement uses the five-second matching rule under the original parser.

Factory	Workers	Videos	Events	Activities	Processes	Coverage	Boundary agreement
001	11	97	2,288	131/125	13/13	94.8%	93.1%
002	10	54	1,541	75/67	24/24	28.2%	71.8%
003	11	72	2,449	55/52	12/12	93.0%	86.8%
004	10	51	1,893	71/65	22/22	92.1%	90.4%
005	11	22	1,103	102/97	18/17	25.0%	75.0%
006	6	26	1,424	61/59	24/24	14.1%	62.1%
All	59	322	10,698			52.8%	82.1%

strict support of $36/42 = 0.857$. The six P cases remain in the denominator, so this rate measures full textual support.

For the generated transcript-entry sample, COV denotes overlap with at least one event, EXC an uncovered entry judged non-operational, and MISS an uncovered entry judged operational. There are 34 COV entries, four MISS entries, and four EXC entries. The reported calculation retains the four uncovered operational entries and excludes only the four uncovered non-operational entries. Dividing the overlapping entries by that retained total gives $34/(34+4) = 0.895$. We call this the *reported entry-overlap rate*, not action recall. One COV entry describes camera tilt covered by a longer event. Thus, the denominator does not establish 38 independently verified operational entries, and overlap does not show that all actions or labels were extracted correctly.

The audit measures consistency with generated transcripts. It cannot identify visual errors, missed actions, or mistaken object interpretations already present in the generated transcripts. There is no independent evaluator, inter-rater agreement estimate, or source-video audit. The sample proportions do not bound error in downstream time shares, category assignments, observation selection, or causal interpretation. Appendix E of the Additional Material reproduces all judgments and their notes, with the unverified truncation explanation clearly marked.

7. Results and Factory Case Studies

7.1. ***RQ1***: *Construction and structural properties*

Table 16 reports the released corpus. All 10,698 events have the required values, nonnegative ordered endpoints, and labels in the relevant factory lists. Start times are nondecreasing within each log. There are no empty logs or zero-duration events. However, 35 adjacent event pairs overlap, 20 in factory **001** and 15 in factory **002**. Passing schema and vocabulary checks therefore does not imply a non-overlapping segmentation. Activity vocabulary use ranges from 89.3% to 96.7%.

7.2. ***RQ2***: *Consistency with generated transcripts*

The original parser finds 82.1% of event boundaries within five seconds of a recognized generated transcript endpoint. It yields an overall coverage diagnostic of 52.8%. Neither statistic is video accuracy. The coverage denominator is heavily affected by the literal multi-hour endpoints listed in Table 12. Boundary agreement can also be low when formats change or when **S4** splits a described interval. These diagnostics alone do not establish extraction truncation.

Table 17. Two separate audit samples against generated transcripts: 42 event-support judgments and 42 transcript-entry-overlap judgments. Strict support divides 36 fully supported events by all 42 sampled events. Entry overlap divides 34 overlapping entries by the 38 retained entries, including four uncovered operational entries and excluding four uncovered non-operational entries.

Sample outcome	Count	Interpretation
Event fully supported (C)	36	Strict support from generated text: 36/42, or 85.7%
Event partially supported (P)	6	Context is plausible; overlapping description is insufficient
Event contradicted (X)	0	No contradiction among the sampled events
Transcript entry overlaps (COV)	34	At least one event overlaps; action completeness not established
Uncovered, non-operational (EXC)	4	Excluded from the archive's reported overlap denominator
Uncovered, operational (MISS)	4	Reported entry-overlap rate: 34/38, or 89.5%

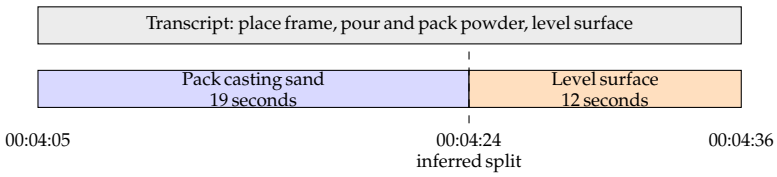


Figure 5. Recorded transcript interval and inferred event intervals for the foundry example. The generated transcript entry is line 16 of its text file; the events are data rows 17 and 18 of the corresponding CSV. Bar lengths show recorded durations, not independently checked action boundaries.

Table 17 summarizes the archived judgments. Partial-support examples show the difficulty of checking events against compressed text. Event sample 5 assigns a particular bagging activity where the generated transcript only states that repetitive assembly and packaging continue. Sample 40 assigns guiding an upper mold half where the text only says that the worker helps with a mold. Transcript-entry sample 10 uses an interval from 11:51:00 to 12:59:00 in an anomalous generated transcript whose event log ends at 00:20:00. Its MISS judgment is preserved, but the original explanation as long-recording truncation is unverified. These cases show why a single accuracy label would obscure different sources of uncertainty.

7.3. An end-to-end foundry example

This example shows how a description of several actions becomes two labeled events with an inferred internal boundary. In factory **006**, worker 003, recording 00003, the generated transcript from **S1** describes actions from 00:04:05 to 00:04:36: placing an empty frame on a pattern plate, pouring and hand-packing powder, and using a wooden tool to level the surface. The worker summary from **S2** provides context by describing preparation, filling, packing, leveling, and compression of molds across recordings. The factory vocabulary from **S3** supplies process *Sand mold creation* and activity *level powder surface with wooden scraper*. The **S3** report also supplies a compact summary of this worker's molding work.

Event extraction in **S4** creates two intervals within that 31-second generated transcript entry: *pack casting sand into mold frame*, from 00:04:05 to 00:04:24, and *level powder surface with wooden scraper*, from 00:04:24 to 00:04:36. Both have process *Sand mold creation*. Figure 5 shows what is retained and added. The internal boundary at 00:04:24 is inferred; **S1** does not timestamp that transition separately. The events compress placement, pouring, and joint hand-packing into the first activity and add shared labels and a process context.

```
{"events": [{
  "estimated_start_time": "00:04:24",
  "estimated_end_time": "00:04:36",
  "activity": "level powder surface with wooden scraper",
  "process": "Sand mold creation"
}]}
```

Table 18. Reproduced time analysis on 320 logs. Left: percentage of summed event duration. Right: medians across eligible workers after the original worker-level rounding. Trans./h means VA-to-non-VA transitions per summed event-hour. Mean VA run is the median of worker mean VA-run durations in minutes. A run sums consecutive same-category rows within a clip; transition counting can cross clips.

	Process-based category shares (%)							Medians across eligible workers		
Factory	VA	MH	TR	QI	CM	RW	DOC	VA (%)	Trans./h	Mean VA run (min)
001	93.1	4.1	1.8	0.0	0.2	0.0	0.9	94.5	9.2	4.6
002	96.3	0.1	2.3	0.2	0.0	0.0	1.1	97.3	3.9	8.8
003	82.1	0.4	0.0	0.0	6.3	10.8	0.4	86.4	3.6	9.0
004	89.9	3.0	3.5	2.0	0.9	0.0	0.7	91.3	10.6	3.6
005	80.6	5.4	9.7	4.3	0.0	0.0	0.0	85.9	5.6	6.2
006	74.2	12.1	7.8	0.0	5.7	0.0	0.3	87.0	14.4	3.1

Listing 6: Illustrative JSON reconstructed from CSV data row 18. It is not an archived model response.

The linked generated transcript, worker summary, factory report, and event log preserve the successive representations of this recording. **S5** retains the recording identifier and local offsets alongside shifted timestamps. Audit event 41 judges the leveling event fully supported by its generated transcript. This supports the label's textual basis, not the exact internal split or a video-based duration measurement.

*7.4. **RQ3**: Descriptive analytical capabilities*

7.4.1. Time composition and category transitions

We summarize the time assigned to each category and the pattern of VA runs for eligible workers. Table 18 separates pooled duration shares from medians across workers.

Table 18 reproduces the existing category shares using the two-log exclusion rule. For factory **006**, 74.2% pools the durations of all retained events, whereas 87.0% is the median of the eligible workers' individual VA shares. Combined shares are calculated before rounding, so they may differ slightly from the sum of the displayed percentages. For example, the combined MH/TR share for factory **006** is 19.8%, while its separately rounded components are 12.1% and 7.8%.

The non-VA category share ranges from 3.7% to 25.8% of retained summed event duration. This is neither total worker time nor recoverable capacity. Factory **003** assigns 10.8% to defect patching. Its factory report describes defect patching in the opening overview and vocabulary. The duration share quantifies this recorded behavior under an explicit category rule.

Figure 6 displays the category shares. The mean VA-run column is the median across workers of their *mean* VA-run durations. A value of 3.1 minutes in factory **006** is a central tendency; it says nothing about the longest VA episode. Factory **001** has 449 MH/TR rows, a mean row duration of 14.54 seconds, and 14.6 such rows per summed event-hour. Factory **006** has 192 MH/TR rows, a mean of 25.14 seconds, and 28.4 rows per summed event-hour. These are row-based descriptions. Segmentation, repeated-action compression, and clip joins influence their values.

7.4.2. Process flow and discovery

Video-as-case traces describe observation windows. Every recording has a distinct activity sequence in factories **001**, **004**, **005**, and **006**. Factory **002** has 53 distinct sequences across 54 recordings, and factory **003** has 70 across 72. Self-loops account for 4.3% to

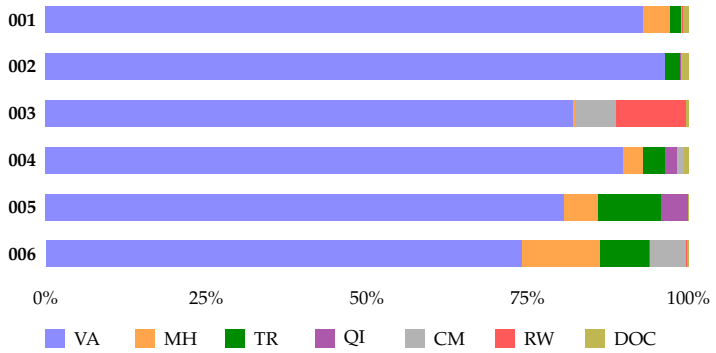


Figure 6. Category composition calculated from retained event rows and the released process map. Segments use unrounded duration totals; Table 18 reports rounded percentages. Factories have different task mixes and label schemes, so the plot does not rank efficiency.

Table 19. Selected shared-dominant-process groups, using the original eligibility and rounding rules. Ranges summarize worker profiles; they are not estimates of individual efficiency.

Factory	Dominant process	Workers	VA share	Mean VA run
001	Manual mechanical assembly	6	89.7 to 97.8%	1.7 to 6.7 min
001	Metal stamping	3	90.6 to 98.5%	3.2 to 7.9 min
003	Surface finishing and polishing	5	67.3 to 96.9%	3.3 to 13.7 min
004	Garment ironing	2	52.8 to 77.1%	3.5 to 4.1 min
004	Overlock seaming	4	87.8 to 96.5%	1.5 to 10.3 min
005	Manual lathe machining	2	95.0 to 95.8%	6.9 to 9.4 min

16.8% of within-video directly-follows pairs. These statistics are reproduced directly from the stored labels. A self-loop can reflect separately extracted repetitions or segmentation choices; it is not a count of manufactured items.

The released discovery script uses Inductive Miner [32] with noise threshold 0.2, video cases, and activity labels. Appendix C of the Additional Material records the repeated calculation and current dependency version. The recalculated fitness ranges from 0.9544 to 0.9910 and precision from 0.0603 to 0.2679. Together, these results show that the discovered models can replay most behavior in the inferred logs while also permitting substantial additional behavior. Fitness measures how the discovered model can replay the inferred traces; precision measures how much additional behavior it permits relative to that log. Both are properties of a model-log pair, not accuracy against actual factory behavior.

Many distinct traces and low precision can arise from flexible work, recording boundaries, abstraction choices, label variation, or extraction mistakes. The present evidence does not separate these causes. No process-level discovery result is reported because its computation and saved output are unavailable. Replacing activity labels with process labels would change the represented behavior and evaluation target. A higher precision after that change would not by itself be a controlled demonstration of better extraction or modeling.

7.4.3. Worker profiles and shared dominant processes

A shared vocabulary permits descriptive comparison within a factory. Table 19 groups workers by their largest process-duration total, then compares VA shares over *all* retained events for those workers. Eligibility requires at least 1,800 seconds of summed event duration. It does not require verified footage duration or equivalent tasks. There are 11, 10, 10, 10, 5, and 5 eligible workers in factories **001** to **006**.

The 29.6 percentage-point range in the surface-finishing group describes different observed profiles. Differences may reflect task mix, product requirements, recording selection, category assignment, or working conditions. Likewise, the small range for two

lathe workers cannot rule out differences in behavior. Neither result supports individual performance rankings.

7.5. Six descriptive factory cases

Each case below states its recorded scope, quantitative result, relationship to the report, and unresolved interpretation. Factory descriptions come from the generated reports; the logs quantify behavior described there. Different vocabularies, products, task mixes, and observation windows limit comparisons. The cases describe recorded work and motivate further study, but do not establish efficiency or the benefits of an intervention.

Factory **001**: hardware stamping and assembly.

The archive has 97 clips from 11 workers; the time analysis retains 95 clips. Manual mechanical assembly accounts for 55.7% of retained summed duration, metal stamping for 18.2%, and component bagging for 11.3%. The report describes material replenishment, floor storage, and operator transport. The 449 MH/TR rows and their short mean duration motivate examining how materials are supplied to workers. Product mix and compressed repetitions could also explain this pattern.

Factory **002**: garment finishing and packaging.

There are 54 clips from ten workers, all retained by the event-end rule. Fabric folding, garment bagging, and ironing contribute 18.8%, 17.1%, and 13.2% of summed duration. VA is 96.3%. The report describes transport, floor staging, and varied folding methods. The small separate handling and transport share may leave those actions counted within other processes. Two anomalous generated transcript endpoints and mixed formats weaken temporal interpretation. Product-specific finishing requirements and category boundaries need closer examination.

Factory **003**: surface preparation and assembly.

The archive covers 72 clips from 11 workers; all clips are retained. Surface finishing and polishing contributes 42.1%, wax pattern assembly 19.2%, and defect patching 10.8%. The report identifies patching and material transport. The log quantifies patching's share without establishing defect rates or avoidable rework. Its 101 transport events total 1,230 seconds, yet the process-based TR share is zero because their durations enter other categories. Material types and quality criteria remain uncertain in the report itself.

Factory **004**: garment sewing and finishing.

There are 51 clips from ten workers, all retained. Overlock seaming contributes 23.5%, garment folding 11.0%, and thread trimming 10.8%. The median worker VA-transition rate is 10.6 per summed event-hour. The report describes tool switching, material handling, and inspection. The logs add a category-transition measure and a 24.3 percentage-point VA-share range in the two-worker ironing group. Product specifications, additional finishing duties, and recording windows could explain this difference.

Factory **005**: machining and motor assembly.

The archive contains 22 clips from 11 workers, but only five workers meet the time threshold. All clips are retained, so the worker medians describe a smaller subset than the factory duration shares. Manual lathe machining contributes 32.1%, drill press machining 19.1%, and material transport and logistics 9.7%. The report describes staging, fixturing, inspection, and coupled workstations. The log assigns 4.3% to dimensional inspection. The

Table 20. Evidence-linked hypotheses. These are candidates for investigation; intervention effects were not measured.

Observation	Plausible explanation	Competing explanation	Evidence needed
001: 449 short MH/TR rows; 5.9% of summed duration	Material presentation creates frequent replenishment	Segmentation splits one handling episode into several rows; task mix differs	Video checks, supply distances, batch quantities, station observations
006: 19.8% MH/TR; frequent category transitions	Layout or replenishment arrangements require repeated movement	Heavy or fragile materials require necessary support work	Routes, loads, safety requirements, matched task observations
003: 10.8% defect patching	Upstream defects require corrective work	Patching is a planned finishing requirement	Product specifications, defect criteria, item counts, upstream records
003 and **004**: VA shares differ within dominant-process groups	Conditions or work allocation differ	Product mix, selection, or extraction differences	Comparable products and observation windows; independent video review
005: two similar lathe VA shares	Common station requirements shape profiles	Similar aggregate shares hide different tasks or behavior	Task-level comparisons and production context

Table 21. Possible meanings of an apparent zero, ordered along the evidence path. Only the last mechanism is directly demonstrated by the factory **003** example; the others are possible mechanisms implied by the pipeline.

Mechanism	Where information is lost	Needed check
Behavior absent from the recorded window	Observation selection	Define the window and verify that it is suitable for the question
Behavior present but not narrated	Video interpretation or narration	Independent review of source footage
Behavior narrated but lost in summaries	**S2** or **S3** compression	Compare generated transcripts with worker and factory summaries
No suitable vocabulary label	Vocabulary induction	Inspect lists against the behavior of interest
Suitable label exists but is not assigned	Event extraction	Compare relevant generated transcript spans and extracted labels
Event exists but category aggregation hides it	Analytical mapping	Compare activity-level rows with process-derived categories

two lathe profiles have similar VA shares, but their task equivalence is unknown. One generated transcript also has an unresolved endpoint anomaly.

Factory **006**: molding and foundry work.

There are 26 clips from six workers, all retained; five workers meet the time threshold. Sand mold creation, pattern cleaning and coating, and compression molding contribute 17.4%, 14.0%, and 12.2%. The report describes floor staging, material movement, cleaning, and mold-track operations. The logs assign 19.8% of summed duration to MH/TR and yield a median worker mean VA run of 3.1 minutes. These quantities support closer study of handling requirements. Necessary mold preparation, differing materials, layout, and segmentation choices could all contribute. Two generated transcript endpoints require resolution against source metadata.

7.6. From observations to operational hypotheses

Table 20 connects observations to explanations and the evidence needed to distinguish them. Reports identify phenomena; logs quantify them under a chosen representation. Differences in task mix, observation windows, and extraction choices remain competing explanations. Establishing causes or comparing efficiency requires matched operational evidence. Transport, inspection, cleaning, and documentation are not automatically waste, and their time is not automatically available for production. No intervention effect or cost saving is demonstrated by these descriptive results.

Table 22. Transformation, possible imperfection, analytical consequence, and existing evidence. A prompt or a permitted operation establishes a possible mechanism, not its error frequency.

Transformation	Possible imperfection	Downstream consequence	Existing evidence
S1: narration	Omitted actions or compressed repetitions; inaccurate times	Row frequency differs from cycle frequency; duration can be distorted	Repetition rule in prompt; seven endpoint anomalies in Table 12; visual omissions unmeasured
S2 and **S3**: summaries and classes	Rare details lost; concepts merged or separated	Vocabulary may omit a needed distinction	Compression requested by prompts; lists available for inspection; information loss not quantified
S4: label assignment	Episode forced into the closest permitted label	Membership passes despite weak textual support	Six partially supported audit events, including the bagging example; mechanism not isolated by the audit
S4: interval assignment	Internal boundary added without a timed transition in the text	Duration attributed to each action depends on the split	Foundry split at 00:04:24; 35 adjacent overlaps in the full archive
S5: consolidation	Filename joins treated as continuous behavior	Added transitions and synthetic elapsed time	Merge code and preserved local offsets; real inter-clip gaps unknown
Analysis: category mapping	Recorded activity hidden by its process category	Apparent zero for a behavior present in the log	Factory **003** has 101 transport events but no process mapped to TR

8. Measurement and Analytical Implications

8.1. Observability and the interpretation of zeros

A zero in an analytical table has several possible causes (Table 21). These causes occur at different stages and require different checks. Requiring a label for transport or waiting improves representability, but does not establish reliable detection or a measured absence. A model may still omit the action, assign another label, or absorb it into a longer episode.

In factory **003**, both *transport loaded tray to storage rack* and *transport bulk material via wheelbarrow* are permitted activity labels and occur in the logs. Together they account for 101 events and 1,230 seconds, yet the TR-category share is zero. Transport remains in the event log, but its duration is counted under other categories because the rule uses broader process labels and no process in this factory maps to TR. The zero describes the category map's output, not an absence of transport or a missing activity label.

To interpret an observed zero as evidence of absence within a window, an analyst would need a defined observation scope, a suitable representation, adequate detection, correct assignment, and an aggregation rule that preserves the behavior. None follows from label membership alone. Conversely, an observed event can establish what the representation contains while leaving the event's visual accuracy unmeasured.

8.2. Uncertainty propagation and limits of provenance

Several generated outputs can agree because they inherited the same earlier mistake. A mistaken object description in **S1** can shape the worker summary, the factory vocabulary, and the final event label along both branches of Figure 2. Provenance helps trace how an event was constructed, so readers can inspect these dependencies. Table 22 links the possible errors to their analytical consequences and the available evidence.

Quality-aware construction [9] and event-log imperfection patterns [23] establish the broader concern. The table separates observed examples from possible mechanisms; it does not estimate how often each error occurs.

These mechanisms occur within an observation scope that is itself incomplete: the selection protocol and source durations are unavailable. No stage-wise error rates are calibrated, and structural checks cannot quantify visual errors. Worker and factory differences may therefore reflect observation selection as well as representational choices.

Readable text is useful evidence, but it is weaker than structured event provenance. Existing records identify a source file and local interval. They do not store supporting

generated transcript-span identifiers, alternative labels, calibrated confidence, or a flag that distinguishes an inferred internal boundary from a copied generated transcript endpoint. Finding the source of an event therefore requires reading and matching intervals. The foundry example can be traced manually, yet its CSV does not record why the split at 00:04:24 was chosen.

A future schema could add generated transcript-entry references, a boundary-origin field, and vocabulary and prompt versions. Data-quality annotations could retain known issues for downstream analysis [24]. Explicit alternative labels or bounded times would instead extend the event representation toward uncertain data [25,33]. Such alternatives need evidence: the five-second matching tolerance does not imply that an event boundary lies within five seconds of its true value. Calibrated confidence values are unavailable for the existing events. Provenance can explain how a conclusion was reached; it does not certify that every preceding observation was correct.

8.3. Missing time does not imply a one-sided bound

The observed share outside VA is sometimes described as a lower bound because waiting and other supporting work can be omitted. That interpretation also assumes that omitted time is sufficiently non-VA. Missing VA time can instead raise the observed non-VA share. We consider the two extreme possibilities: all missing time is VA, or all missing time is non-VA.

Proposition.

Consider an idealized observation with $C > 0$ seconds of correctly classified, non-overlapping observed time, of which N seconds are non-VA. Let $U \geq 0$ seconds be unobserved. If observed and unobserved time partition a known target window, then its full-time non-VA share satisfies

$$\frac{N}{C+U} \leq p_{\text{nonVA}} \leq \frac{N+U}{C+U}. \tag{4}$$

Proof. Let x be the non-VA duration in the unobserved part. Since $0 \leq x \leq U$, the full share is $(N+x)/(C+U)$ and has the stated extrema. Both are attainable: all missing time may be VA or all may be non-VA.

The lower endpoint assigns all missing time to VA; the upper endpoint assigns all of it to non-VA. For $U > 0$, the observed ratio N/C is a lower bound only if $x/U \geq N/C$: the missing time must contain at least the observed proportion of non-VA time. Thus, merely knowing that some waiting is missing is insufficient. The interval width in Equation 4 is $U/(C+U)$, the missing fraction of the target window. It shows how unknown coverage limits a share even with perfect observed labels.

As an illustration, suppose 80 seconds are observed, 20 of them are non-VA, and 20 seconds are missing. The observed non-VA share is 25%. Over the full 100-second window, it could be as low as 20% if all missing time is VA, or as high as 40% if all missing time is non-VA.

This is a conceptual result, not an empirical confidence interval for this dataset. Source durations and therefore U are unavailable; labels are inferred; and summed event durations include overlaps. Substituting the coverage diagnostic or the 97.1 retained summed event-hours into this proposition would violate its assumptions. The result identifies the additional evidence needed before making a bound claim.

8.4. Why aggregation changes the question

A duration-weighted factory share estimates the composition of the retained event rows. A median worker share gives equal weight to each eligible worker. They answer different questions, as the factory **006** example beside Table 18 shows. Different observation amounts and work profiles can produce this difference without either statistic being erroneous.

Likewise, process-based and activity-based categories need not agree. Counting rows emphasizes the extraction granularity; counting runs suppresses adjacent labels; measuring interval unions removes overlap but does not resolve classification conflicts. None is a universal correction. A defensible analysis states its unit, denominator, case rule, handling of gaps, and interpretation before drawing operational conclusions. The released measurements serve as a reproducible example of those choices and their limits.

9. Discussion and Conclusions

9.1. Main findings

Egocentric recordings can be organized into a shared representation of manual work through generated descriptions, shared factory labels, and interval-event extraction. For **RQ1**, the released logs satisfy the reported field, ordering, and vocabulary checks. For **RQ2**, 36 of 42 sampled events were fully supported by their generated descriptions, while accuracy against the videos remains unmeasured. These results distinguish structural consistency and textual support from correct observation.

For **RQ3**, the logs support descriptions of work composition, local sequences, and worker profiles. They quantify behaviors described in the reports and identify questions for further investigation. For **RQ4**, the examples show how representation choices affect these measurements. In factory **003**, the process map counts recorded transport under other categories. In the foundry example, an inferred boundary divides a described episode between activities. Synthetic clip joins can add transitions, and missing time can move a category share in either direction. Interpreting a result therefore requires its category rule, event boundaries, case definition, and denominator.

9.2. Limits and responsible use

The vocabulary is induced from the same factory corpus used for assessment, so the results do not establish transfer to unseen factories. The single-author audit uses generated text and cannot reveal visual errors shared by the generated transcripts and logs.

Missing source durations prevent us from measuring footage coverage or the amount of missing time. Missing selection records prevent us from establishing how representative the processed recordings are. Missing historical execution details prevent exact regeneration of the model outputs. The released artifacts still permit reproduction of the descriptive calculations.

Different task mixes, observation windows, and category assignments can explain differences between worker profiles. Product identifiers and production outcomes are absent, and no intervention was evaluated. The results consequently support hypotheses about process design, not causal diagnoses, efficiency rankings, or estimates of recoverable capacity.

Worker identifiers are pseudonymous and may still enable identification when combined with local information [34]. Use should remain tied to an agreed analytical purpose, with suitable aggregation and review by the people whose work is represented. These inferred profiles do not support employee appraisal, pay, or disciplinary decisions. The source

card does not document the original consent procedure; this secondary study collected no new consent.

9.3. Next steps

The linked generated transcripts, summaries, vocabularies, and logs provide a basis for testing alternative event representations. Priorities are independent video assessment with multiple evaluators, preservation of source durations, and explicit links from events to supporting generated transcript spans. Product references and real calendar times would be needed to extend the resource to synchronized production-flow analysis.

Additional Material

The accompanying Additional Material contains the evidence and assessment scope (Appendix A), reproducibility information and configuration (Appendix B), activity-level discovery results (Appendix C), the complete operational-category map (Appendix D), and all archived audit judgments (Appendix E). The four full prompt templates appear in Listings 2–5 in Section 4.

Declarations

Author Contributions: Alessandro Berti carried out the experiments. Wil M. P. van der Aalst reviewed the manuscript and suggested improvements.

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Institutional Review Board Statement: This study is a secondary analysis of the existing Egocentric-10K dataset. It involved no recruitment of participants, intervention, or new video collection. No ethics approval or exemption record for this secondary analysis is reported.

Informed Consent Statement: No participants were recruited or contacted for this secondary analysis. The source videos were obtained from the existing Egocentric-10K dataset. The authors did not collect new consent, and the source dataset card does not report the original consent procedure.

Data Availability Statement: The pipeline, prompts, generated transcripts, factory vocabularies, derived event logs, and assessment code are available at https://github.com/fit-alessandro-berti/annotated-egocentric-10k-dataset. The source videos are available from Build AI's Egocentric-10K dataset at https://huggingface.co/datasets/builddotai/Egocentric-10K, under its access terms and Apache License 2.0. The repository contains derived artifacts and does not redistribute the source videos.

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Conflicts of Interest: Wil M. P. van der Aalst is affiliated with Celonis, a process mining software company. No other conflicts of interest are reported.

References

1. van der Aalst, W.M.P. *Process Mining: Data Science in Action*, 2 ed.; Springer, 2016. https://doi.org/10.1007/978-3-662-49851-4.

2. Kratsch, W.; König, F.; Röglinger, M. Shedding light on blind spots - Developing a reference architecture to leverage video data for process mining. *Decis. Support Syst.* **2022**, *158*, 113794. https://doi.org/10.1016/j.dss.2022.113794.
3. Build AI. Egocentric-10K: Dataset Card. https://huggingface.co/datasets/builddotai/Egocentric-10K, 2025. Accessed 19 October 2026; current revision 3e5f87c.
4. Gavric, A.; Bork, D.; Proper, H. Multimodal Process Mining. In Proceedings of the 2024 IEEE 26th Conference on Business Informatics (CBI). IEEE, 2024, pp. 99 to 108. https://doi.org/10.1109/CBI62504.2024.00021.
5. Petersen, N.; Pagani, A.; Stricker, D. Real-Time Modeling and Tracking Manual Workflows from First-Person Vision. In Proceedings of the 2013 IEEE International Symposium on Mixed and Augmented Reality (ISMAR). IEEE, 2013, pp. 117 to 124. https://doi.org/10.1109/ISMAR.2013.6671771.
6. Pham, T.; Xiao, Y. Unsupervised Workflow Extraction from First-person Video of Mechanical Assembly. In Proceedings of the 19th International Workshop on Mobile Computing Systems and Applications. ACM, 2018, pp. 31 to 36. https://doi.org/10.1145/3177102.3177112.
7. König, F.; Egger, A.; Kratsch, W.; Röglinger, M.; Wördehoff, N. Unstructured Data in Process Mining: A Systematic Literature Review. *ACM Trans. Manag. Inf. Syst.* **2025**, *16*, 25:1 to 25:34. https://doi.org/10.1145/3727148.
8. Diba, K.; Batoulis, K.; Weidlich, M.; Weske, M. Extraction, correlation, and abstraction of event data for process mining. *WIREs Data Mining Knowl. Discov.* **2020**, *10*, e1346. https://doi.org/10.1002/widm.1346.
9. Andrews, R.; van Dun, C.G.J.; Wynn, M.T.; Kratsch, W.; Röglinger, M.; ter Hofstede, A.H.M. Quality-Informed Semi-Automated Event Log Generation for Process Mining. *Decis. Support Syst.* **2020**, *132*, 113265. https://doi.org/10.1016/j.dss.2020.113265.
10. Knoch, S.; Ponpathirkoottam, S.; Schwartz, T. Video-to-Model: Unsupervised Trace Extraction from Videos for Process Discovery and Conformance Checking in Manual Assembly. In Proceedings of the BPM. Springer, 2020, Vol. 12168, *Lecture Notes in Computer Science*, pp. 291 to 308. https://doi.org/10.1007/978-3-030-58666-9_17.
11. Lepsien, A.; Koschmider, A.; Kratsch, W. Analytics Pipeline for Process Mining on Video Data. In Proceedings of the BPM (Forum). Springer, 2023, Vol. 490, *Lecture Notes in Business Information Processing*, pp. 196 to 213. https://doi.org/10.1007/978-3-031-41623-1_12.
12. Pegoraro, M.; Seng, J.; Heller, D.; van der Aalst, W.M.P.; Kersting, K. All Eyes on the Workflow: Automated and Efficient Event Discovery from Video Streams. *CoRR* **2026**, *abs/2604.22476*. https://doi.org/10.48550/arXiv.2604.22476.
13. Brzychczy, E.; Aleknonytė-Resch, M.; Janssen, D.; Koschmider, A. Process mining on sensor data: a review of related works. *Knowl. Inf. Syst.* **2025**, *67*, 4915 to 4948. https://doi.org/10.1007/s10115-024-02297-y.
14. Krishna, R.; Hata, K.; Ren, F.; Fei-Fei, L.; Niebles, J.C. Dense-Captioning Events in Videos. In Proceedings of the IEEE International Conference on Computer Vision (ICCV). IEEE, 2017, pp. 706 to 715. https://doi.org/10.1109/ICCV.2017.83.
15. Kecht, C.; Egger, A.; Kratsch, W.; Röglinger, M. Event Log Construction from Customer Service Conversations Using Natural Language Inference. In Proceedings of the 2021 3rd International Conference on Process Mining. IEEE, 2021, pp. 144 to 151. https://doi.org/10.1109/ICPM53251.2021.9576869.
16. Shirali, M.; Fani Sani, M.; Ahmadi, Z.; Serral, E. LLM-Based Event Abstraction and Integration for IoT-Sourced Logs. In Proceedings of the Business Process Management Workshops. Springer, 2025, Vol. 534, *Lecture Notes in Business Information Processing*, pp. 138 to 149. https://doi.org/10.1007/978-3-031-78666-2_11.
17. Berti, A.; Kourani, H.; van der Aalst, W.M.P. PM-LLM-Benchmark: Evaluating Large Language Models on Process Mining Tasks. In Proceedings of the ICPM Workshops. Springer, 2025, Vol. 533, *Lecture Notes in Business Information Processing*, pp. 610 to 623. https://doi.org/10.1007/978-3-031-82225-4_45.
18. Dani, V.S.; Dees, M.; Leopold, H.; Busch, K.; Beerepoot, I.; van der Werf, J.M.E.M.; Reijers, H.A. Event Log Extraction for Process Mining Using Large Language Models. In Proceedings of the CoopIS. Springer, 2025, Vol. 15506, *Lecture Notes in Computer Science*, pp. 56 to 72. https://doi.org/10.1007/978-3-031-81375-7_4.
19. Seeth, M.; Tavares, G.M.; Schuster, D. Automated Event Log Generation from Unstructured Text Using Finetuned LLMs, 2026, [2609.01320]. Preprint.
20. van Zelst, S.J.; Mannhardt, F.; de Leoni, M.; Koschmider, A. Event Abstraction in Process Mining: Literature Review and Taxonomy. *Granular Computing* **2021**, *6*, 719 to 736. https://doi.org/10.1007/s41066-020-00226-2.
21. Kapugama Geeganage, D.T.; Wynn, M.T.; ter Hofstede, A.H.M. Text2EL+: Expert Guided Event Log Enrichment Using Unstructured Text. *ACM J. Data Inf. Qual.* **2024**, *16*, 8:1 to 8:28. https://doi.org/10.1145/3640018.
22. Sturm, F.; Liessmann, A.; Matzner, M. Multimodal Event Log Construction for Process Mining: Instantiating a Reference Architecture. In Proceedings of the Design for Better Futures: Beyond the Science of the Artificial. Completed Research. Springer, 2026, Vol. 16605, *Lecture Notes in Computer Science*, pp. 146 to 164. https://doi.org/10.1007/978-3-032-28316-0_9.
23. Suriadi, S.; Andrews, R.; ter Hofstede, A.H.M.; Wynn, M.T. Event Log Imperfection Patterns for Process Mining: Towards a Systematic Approach to Cleaning Event Logs. *Information Systems* **2017**, *64*, 132 to 150. https://doi.org/10.1016/j.is.2016.07.011.

24. Goel, K.; Leemans, S.J.J.; Martin, N.; Wynn, M.T. Quality-Informed Process Mining: A Case for Standardised Data Quality Annotations. *ACM Trans. Knowl. Discov. Data* **2022**, *16*, 97:1 to 97:47. https://doi.org/10.1145/3511707.
25. Pegoraro, M.; Uysal, M.S.; van der Aalst, W.M.P. Conformance Checking over Uncertain Event Data. *Information Systems* **2021**, *102*, 101810. https://doi.org/10.1016/j.is.2021.101810.
26. Esser, S.; Fahland, D. Multi-Dimensional Event Data in Graph Databases. *J. Data Semant.* **2021**, *10*, 109 to 141. https://doi.org/10.1007/s13740-021-00122-1.
27. van der Aalst, W.M.P. Object-Centric Process Mining: Dealing with Divergence and Convergence in Event Data. In Proceedings of the SEFM. Springer, 2019, Vol. 11724, *Lecture Notes in Computer Science*, pp. 3 to 25. https://doi.org/10.1007/978-3-030-30446-1_1.
28. Corallo, A.; Lazoi, M.; Striani, F. Process Mining and Industrial Applications: A Systematic Literature Review. *Knowledge and Process Management* **2020**, *27*, 225 to 233. https://doi.org/10.1002/kpm.1630.
29. Sener, F.; Chatterjee, D.; Shelepov, D.; He, K.; Singhania, D.; Wang, R.; Yao, A. Assembly101: A Large-Scale Multi-View Video Dataset for Understanding Procedural Activities. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022, pp. 21096 to 21106.
30. Gebru, T.; Morgenstern, J.; Vecchione, B.; Vaughan, J.W.; Wallach, H.; Daumé III, H.; Crawford, K. Datasheets for Datasets. *Communications of the ACM* **2021**, *64*, 86 to 92. https://doi.org/10.1145/3458723.
31. Berti, A.; van Zelst, S.J.; Schuster, D. PM4Py: A process mining library for Python. *Softw. Impacts* **2023**, *17*, 100556. https://doi.org/10.1016/j.simpa.2023.100556.
32. Leemans, S.J.J.; Fahland, D.; van der Aalst, W.M.P. Discovering Block-Structured Process Models from Event Logs - A Constructive Approach. In Proceedings of the Petri Nets. Springer, 2013, Vol. 7927, *Lecture Notes in Computer Science*, pp. 311 to 329. https://doi.org/10.1007/978-3-642-38697-8_17.
33. Pegoraro, M.; van der Aalst, W.M.P. Mining Uncertain Event Data in Process Mining. In Proceedings of the 2019 International Conference on Process Mining. IEEE, 2019, pp. 89 to 96. https://doi.org/10.1109/ICPM.2019.00023.
34. Mannhardt, F.; Petersen, S.A.; Oliveira, M.F. Privacy Challenges for Process Mining in Human-Centered Industrial Environments. In Proceedings of the Intelligent Environments. IEEE, 2018, pp. 64 to 71. https://doi.org/10.1109/IE.2018.00017.